

Trustworthy Machine Learning Assignment 1

Team number - 65 (Team LV)

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Membership Inference Attack using LiRA and Confidence-Based Scoring

Membership Inference Attacks (MIAs) main goal is to determine if a data sample was used to train a machine learning model. In this assignment, we analyze a pre-trained ResNet-18 and try to infer membership for unseen images.

The attacker is given a trained ResNet-18 model, a public dataset with known membership labels, and a private dataset where membership is unknown. The goal is to assign a score between 0 and 1 for each sample, where higher values indicate a higher chances of being a training member.

Methodology

We apply a Likelihood Ratio Attack (LiRA) and enhance it by incorporating confidence score fusion -

A) Confidence-Based Attack

We first use the pretrained model to compute prediction probabilities for each input image. For each sample, we extract the maximum softmax probability (confidence score):

- High confidence means model is more certain which further implies it was likely seen during training
- Low confidence means model is uncertain which implies likely unseen

These confidence values are later converted into a $[0,1]$ range to make them comparable across samples.

B) LiRA (Likelihood Ratio Attack)

We implement a LiRA (Likelihood Ratio Attack) using multiple shadow models.

Shadow Model Training

We train 64 shadow models, each on a random subset of the public dataset to simulate the target model's behavior. For each model, we compare losses on training (member) data versus unseen (non-member) data, including private samples.

The main idea is that training samples usually have lower loss and higher confidence than unseen ones. Based on this, we compute a LiRA score by comparing the loss distributions of member and non-member data.

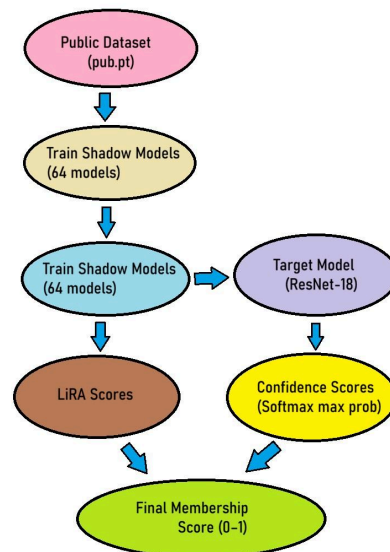
This captures the difference in model behavior more effectively than using confidence alone, making it easier to distinguish training data from non-members.

C) Score Combination

The final membership score is computed by combining the LiRA score and the confidence score, with equal weight given to both. Before combining, each score is normalized so they are on the same scale.

$$\text{Final Score} = 0.5 \cdot \text{LiRA} + 0.5 \cdot \text{Confidence}$$

This combination makes the attack more robust, since it captures both the model's confidence and the differences in how it behaves on training versus unseen data.



Evaluation Metric

TPR @ 5% FPR on pub.pt = 0.0587

This calculates how many true members are correctly identified while keeping false positives low.

Conclusion

- Deep learning models can unknowingly leak information about their training data through prediction behavior.
- It is possible to decide whether a sample was part of the training set without direct access to the data.
- This is done using model confidence, shadow model behavior, and differences in loss values.
- Combining LiRA with confidence-based scoring improves attack strength and stability.
- LiRA helps better separate member and non-member samples, while shadow models improve generalization of signals.
- The results highlight serious privacy risks, especially for sensitive datasets.

Github Link

https://github.com/Ananya-Bhardwaj/TML26_A1_65